

Policy update on a **fail-only** trajectory: Vanilla GRPO vs SeRPO

Same failed rollout (final outcome = fail). Each column walks reward → advantage $\hat{A}_i$ → PPO effect on π .

shared trajectory: s_1 explore · s_2 fetch · **s_3 premature action (fails)** · s_4 retry

X Vanilla GRPO (outcome reward)

1 · REWARD — ONE SCALAR FOR THE WHOLE TRAJECTORY

s_1	$r = 0$
s_2	$r = 0$
s_3	$r = 0$
s_4	$r = 0$

↓ group-normalize: $\hat{A} = (r - \text{mean})/\text{std}$

2 · ADVANTAGE $\hat{A}_i$ — ALL ROLLOUTS FAILED → STD = 0

s_1	0
s_2	0
s_3	0
s_4	0

↓ PPO: $\nabla \propto \hat{A} \cdot \nabla \log \pi$

3 · EFFECT ON POLICY π

s_1	—
s_2	—
s_3	—
s_4	—

$\hat{A} = 0$ on every token → gradient = 0 → **no parameter update**. The failing step s_3 is never down-weighted; the model repeats the same mistake next time.

π unchanged — nothing learned

✓ SeRPO (segment-level reward)

1 · REWARD — ONE PER SEGMENT

s_1	5
s_2	4
s_3	1
s_4	4

↓ normalize within trajectory: $\hat{A}$ varies

2 · ADVANTAGE $\hat{A}_i$ — DIFFERS ACROSS SEGMENTS

s_1	+
s_2	+
s_3	—
s_4	+

↓ PPO: $\uparrow \pi$ if $\hat{A} > 0$, $\downarrow \pi$ if $\hat{A} < 0$

3 · EFFECT ON POLICY π

s_1	$\pi \uparrow$
s_2	$\pi \uparrow$
s_3	$\pi \downarrow$
s_4	$\pi \uparrow$

$\hat{A} \neq 0$ within the trajectory → gradient $\neq 0$. PPO **raises** π for the segments that worked and **lowers** π for the failing step s_3 — a targeted update even though the trajectory failed.

π updated — failure step s_3 down-weighted